

Research Article

Quantifying microbial and harmful algal bloom interactions dynamics through machine learning predictive frameworks

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ABSTRACT

Harmful algal bloom (HAB) mitigation using microorganisms faces challenges, like inconsistent algicidal efficacy, scalability issues, limited microbial diversity, and traditional modeling constraints. To address these challenges, for the first time, this study investigates the quantitative impacts of microorganisms on HABs using machine learning (ML). We conducted an analysis of over 100 microbial species, laying the foundation to optimize treatments for more consistent outcomes. Our findings showed that these microorganisms exhibited an average algicidal ratio of 71 % against HABs. Random Forest, Bagging, and Extreme Gradient Boosting Regressors ML models demonstrated robust predictive performance for algicidal ratios. Nonlinear relationships with time, microbial concentrations, and algal biomass were observed. Optimal culture time for maximizing algicidal ratios ranged from 7 to 10 days; extending time did not enhance the algicidal process. The optimum concentration of microorganisms was identified as 0.5×10^{10} cells/mL, with higher concentrations showing diminishing returns. Additionally, algal biomass influenced the algicidal process, with a sharp decline in algicidal ratio at low biomass levels ($0-1 \times 10^7$ cells/mL), followed by stabilization at higher densities (1.5×10^7 cells/mL). Algal biomass, culture time, *Brevibacillus*, microbial concentration, *Enterobacter*, *Microcystis aeruginosa*, and *Alexandrium tamarense* emerged as crucial factors driving microbial algicidal processes against HABs. Overall, ML analysis of microbial impacts on HABs guides effective treatment optimization with future work emphasizing real-world field applications.

1. Introduction

Harmful algal blooms (HABs) deplete oxygen levels in natural water systems, produce unpleasant odors, and contribute to significant biodiversity loss and ecological damage (Alvarez Dalinger et al., 2024; Cheng et al., 2024). These blooms are particularly hazardous due to the release of toxic substances into water (Zhou et al., 2024). These toxins can have devastating effects on a wide range of marine and freshwater organisms including fishes, shellfish, and other aquatic life forms (Grund et al., 2022; Li et al., 2024b).

Notably, these substances can impair the health of co-occurring species leading to disease or death and creating imbalances in populations (Anderson, 2009). In recent years, initiatives to mitigate HABs in eutrophic water have been extensive, with numerous studies evaluating different control mechanisms (Coral et al., 2013; Deng et al., 2017;

Qi et al., 2016).

The advancement of new and effective methods for managing HABs is crucial to ensure the health of aquatic ecosystems and to safeguard public health (Ou-yang et al., 2022). Rectifying HABs involves a multifaceted approach that involves several monitoring methods (Kordahi et al., 2024). One strategy for HAB mitigation involves the use of microorganisms as biological control agents (Park et al., 2024; Yang et al., 2017). This mode of control has gained widespread recognition for its effectiveness in freshwater environments (Pal et al., 2020).

Microorganisms use various algicidal mechanisms, including structural damage to algae, inhibition of photosynthesis, oxidative stress, and interference with gene expression (Li et al., 2024a). The search for naturally occurring biotic agents that show high efficiency, selectivity, and environmental friendliness is ongoing. However, the integration and optimization of these approaches remain in need of significant

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improvement (Li et al., 2024a).

Traditional research on microbial-mediated methods for HAB control has yielded inconsistent results. For instance, despite the fact that some microorganisms have demonstrated algicidal traits and have the potential to effectively mitigate HABs, others play a role in promoting bloom formation (Choi et al., 2005). An added problem associated with biotic HAB mitigation is the large-scale production, storage, and application of microbial methods.

Extrapolating laboratory-verified results into real-world environmental conditions remains a substantial barrier. Significant efforts are needed to bridge the gap between laboratory success and field efficacy, largely due to the differing life cycles of biological organisms (Sun et al., 2018). Moreover, the concentration of microorganisms is a critical consideration, yet our understanding of the underlying mechanisms of interaction is limited. Many cyanobacteria species are yet to be fully explored (Pal et al., 2020).

A significant challenge in the application of biological agents against HABs lies in the absence of standardized criteria for conducting bioassays of natural agents. Critical variables, such as culture time and initial inoculating biomass, play pivotal roles in determining outcomes, yet these factors are often poorly controlled (Chrysai-Tokousbalides et al., 2007; Churro et al., 2009). Additionally, the variable responses of different HAB species to biological agents further complicate the algicidal process.

Prior to practical application, comprehensive assessments are needed to distinguish whether observed effects are strictly algistatic or algicidal, or a mix of both (Shao et al., 2013). Previous research has faced several limitations, such as the narrow sources of bacterial samples, limited microbial diversity, challenges in isolation and screening methods, and insufficient evaluation of algicidal specificity among diverse algal species. Such constraints hinder the comprehensive understanding required for effective microorganisms-based control of HABs (Coyne et al., 2022).

Commonly, one-way analysis of variance (ANOVA) is employed to assess the significance of treatment effects in published research, with results typically reported as means and standard deviations (Liu et al., 2022a). While traditional statistical approaches provide valuable initial insights, they are inadequate for capturing the complex, nonlinear interactions present in large datasets (Zhang et al., 2020). To address the complexities arising from factors such as microbial diversity, nonlinear relationships, and variations in algicidal concentrations during algal lysis, advanced analytical technologies are essential (Demuez et al., 2015).

A more profound insight into the interactions between algae and microorganisms, along with the specific mechanisms of algal cell lysis, is expected to be a crucial breakthrough in reducing the cost of microalgae cell disruption (Demuez et al., 2015). The limitations of conventional methods can be addressed by utilizing data-driven approaches, particularly machine learning, which are capable of analyzing complex datasets and identifying the most influential factors among a variety of variables (Baek et al., 2021).

We employed four robust models, previously screened in our research, to investigate the dynamics of microorganism and algal interactions. In essence, this study employed a machine learning approach to directly forecast algicidal ratios. The specific objectives of this study are: (1) To collect data from various experimental sources to evaluate the performance of previously identified robust models. The data collected include microbial species, experimental groups, tested HAB species, culture time, microorganism concentration, and algal biomass, with algicidal ratio as the target variable. (2) To quantify the dynamics of microorganism–algal interactions. (3) To determine threshold levels for culture time, microorganism concentration, and algal biomass. (4) To identify the most influential factors affecting the algicidal process.

2. Materials and methods

2.1. Data acquisition and preprocessing

We collected a comprehensive dataset comprising 3070 samples and 21,490 data points sourced from peer-reviewed literature, with detailed references provided in the supplementary material (Table S1). This extensive dataset was curated to enhance the robustness and credibility of our analysis, focusing on microbial and HAB interactions. Data extraction was performed directly from tables and figures within the publications using plot digitizer software, following the methodology described by Yang et al. (2023).

The dataset encompassed seven key parameters essential for modeling microbial–HAB dynamics: (1) microbial species, identifying the specific microorganisms under investigation; (2) experimental or control group designation; (3) tested HAB species; (4) culture time, indicating the duration of the experiments; (5) microorganism concentration, quantifying the number of microorganisms used; (6) algal biomass, reflecting the extent of algal growth; and (7) algicidal ratio (%), measuring the effectiveness of the microorganisms in suppressing algal growth.

Data preprocessing is a critical step in the machine learning workflow, significantly improving model performance, generalizability, and interpretability (García et al., 2016). To ensure the reliability and quality of our dataset, we implemented several robust preprocessing procedures. Outliers were first identified and removed using the interquartile range (IQR) method, which measures the spread between the first and third quartiles, encompassing the middle 50 % of the data. Data points falling outside 1.5 times the IQR were excluded to minimize potential skew in the results (Joshi et al., 2024).

Missing values, which accounted for 5 % of the dataset, were imputed using the k-nearest neighbors (KNN) algorithm ($k = 5$), utilizing the similarity of neighboring data points to preserve data integrity (Yu et al., 2021). Three categorical variables, namely microbial species, experimental group, and tested HAB species, were converted into a numerical format using one-hot encoding, enabling the model to capture distinct effects of each category. By applying one-hot encoding, the model can handle species-specific data, with each species treated independently as a separate feature. A similar approach was used to distinguish unique contributions from different reactive halogen species when aggregating a large dataset (Liang et al., 2024).

To ensure uniform scaling across all features, numerical datasets were normalized using min-max normalization, transforming the values to a range between 0 and 1 (Liu et al., 2022b). These preprocessing steps were implemented using Python's scikit-learn library, specifically utilizing the functions SimpleImputer, OneHotEncoder, and MinMaxScaler for imputation, encoding, and normalization, respectively. The preprocessed data was thoroughly verified to guarantee consistency and readiness for model training.

2.2. Model construction and evaluation

To quantify the dynamics of microbial–HAB interactions, we evaluated four robust and widely adopted machine learning models: Random Forest Regressor (RFR), Bagging Regressor (BR), Extreme Gradient Boosting Regressor (XGBR), and Decision Tree Regressor (DTR). Each model was selected for its specific strengths: RFR is highly effective for handling large datasets and mitigating overfitting through ensemble learning; BR enhances model stability and accuracy via bootstrapping; XGBR is well suited for complex datasets due to its advanced gradient-boosting optimization; and DTR provides interpretability through a hierarchical tree structure. These models were tested to assess their predictive accuracy and effectiveness in modeling microbial–HAB interactions (Yan et al., 2022).

Other models such as Support Vector Machines (SVMs) were not considered in this study. Although SVMs perform well with certain datasets, they are less suited in this context due to their computational intensity and sensitivity to hyperparameter tuning, particularly with high dimensional datasets and categorical variables. Additionally, SVMs offer limited interpretability regarding feature importance, which was a critical objective of this research (Cervantes et al., 2020). While neural networks excel in capturing complex patterns in very large datasets, such as images or time series data, our dataset is of moderate size compared to typical deep learning applications (Li et al., 2022b).

The dataset was split into training (80 %) and testing (20 %) subsets to evaluate model performance on unseen data. To ensure robust training and to minimize overfitting, we implemented 10-fold cross-validation, in which the training set was divided into 10 subsets. Each subset served once as the validation set while the remaining nine were used for training. This process was repeated 10 times, and the average performance metrics were computed to assess model reliability (Ai et al., 2023).

Model implementation was carried out in Python (version 3.9) using the scikit-learn and XGBoost libraries (Lyu et al., 2024). The specific functions employed included RandomForestRegressor, BaggingRegressor, XGBRegressor, and DecisionTreeRegressor, with hyperparameter tuning conducted using GridSearchCV to optimize model performance (Lai et al., 2023). GridSearchCV systematically evaluates predefined combinations of hyperparameters using 10-fold cross-validation to identify the optimal configuration that maximizes the coefficient of determination (R^2) on the validation set while minimizing overfitting.

Model performance was quantitatively evaluated using three metrics: coefficient of determination (R^2), mean absolute error (MAE), and root mean square error (RMSE). R^2 measures the proportion of variance in the target variable explained by the model, MAE quantifies the average absolute difference between predicted and actual values, and RMSE assesses the standard deviation of prediction errors. Superior model performance is indicated by R^2 values closer to 1, and lower MAE and RMSE values reflect higher predictive accuracy (Lyu et al., 2024).

2.3. Quantification of microbial–HAB interaction dynamics

The selected models were used to quantify the interaction dynamics between microorganisms and HABs. To achieve a detailed understanding, we utilized Individual Conditional Expectation (ICE) plots, an extension of the Partial Dependence Plot (PDP) framework, which enables the visualization of model predictions at the level of individual observations. This approach allowed us to examine the variability in predictions across different samples for specific predictors.

ICE plots illustrate how the predicted model outcome for each observation changes as the value of a selected predictor varies across its range while all other predictors are held constant (Maloney et al., 2022). Consequently, each observation is represented by a curve demonstrating how its predicted outcome is affected by changes in that predictor.

In addition to ICE plots, PDPs were used to offer a graphical representation of the average effect of a predictor on the model's response, under the assumption of independence between the variable of interest and all other features (Talukdar et al., 2024). PDPs visualize the marginal effect of a predictor by averaging across all observations, resulting in a curve or surface that shows the overall trend. Because PDPs present averaged effects, they are effectively complemented by ICE plots, which reveal individual-level variability.

To further quantify each feature's contribution to the model, we assessed feature importance (Lv et al., 2024). In machine learning, feature importance typically measures the extent to which a predictor reduces error or influences model predictions. This quantitative assessment provides a numerical ranking of predictor influence, complementing the insights gained from ICE and PDP analyses.

3. Results and discussion

3.1. Diversity of microorganisms in HAB control

The aggregation of data from over 100 research articles on which microbial species are effective against HABs makes a comprehensive source for advancing algal bloom biological control approaches. Given the wide diversity of unexplored HABs reported in the literature, this dataset addresses a significant gap by systematically compiling a range of microbial and HAB species from numerous studies, including bacteria, fungi, actinomycetes, and others.

They have been experimentally evaluated for their efficacy against various HABs as detailed in the supplementary file (Table S2). These data not only highlight well-known algicidal microorganisms but also introduce lesser-studied species that show promise for controlling HABs. This preliminary collection paves the way for further exploration and development of microbial-based strategies for HAB mitigation.

3.2. Descriptive statistics

Table 1 presents descriptive statistics of culture time, concentration, algal biomass, and algicidal ratio based on 2500 experimental observations from the literature which used microbial species to control HABs. The culture time ranged from 0 to 25 days, with a mean of 5 days and a median of 3 days. The 25 %, 50 %, and 75 % percentiles of culture time indicate that most experiments were conducted during early to mid-stages of algal growth, encompassing both short-term and long-term effects on these microorganisms.

Microorganism concentrations varied from 0 to 1.00×10^{12} cells/mL, with most experiments conducted at moderate concentrations (25th to 75th percentiles: 1×10^6 to 1×10^8 cells/mL), although a few included higher concentrations. Algal biomass concentration also exhibited high variability, with an average of 2.80×10^7 cells/mL. Moreover, algicidal ratio, a statistical metric indicating the effectiveness of microorganisms, ranged from 30% to 100 % with a mean of 71 %. The 25th, 50th, and 75th percentiles demonstrate consistent effectiveness among microorganisms, with some experiments achieving complete elimination of HABs at 100 %.

The compiled dataset, encompassing 2500 experimental observations, enabled a comprehensive statistical overview of microbial efficacy against HABs. This extensive data, surpassing the limitations of smaller datasets in previous studies, highlights the diverse experimental conditions and overall promising algicidal potential of microorganisms (Guan et al., 2014; Liu et al., 2022a).

The patterns observed in culture time, concentration, algal biomass, and algicidal ratio provide a foundational understanding for future investigations into optimizing microbial treatments. This work underscores the critical need for further detailed research to fully unravel the complex interplay between variables and ultimately refine microbial approaches to mitigate the ecological and economic impacts of HABs (Li et al., 2022a; Lin et al., 2013; Lyu et al., 2019).

Table 1
Statistical summary of dataset ($n = 2500$) assessing the impact of microorganisms on the control of HABs.

Statistical metric	Culture time (Days)	Concentration (Cells/mL)	Algal biomass (Cells/mL)	Algicidal ratio (%)
Mean	5.00	3.5×10^{10}	2.80×10^7	71.00
Std deviation	5.30	1.72×10^{11}	8.06×10^7	21.22
Min	0.00	0.00	0.00	30.00
25 %	1.00	1.00×10^6	1.2×10^5	53.00
50 %	3.00	1.00×10^7	3×10^6	73.50
75 %	7.00	1.00×10^8	5.21×10^7	90.25
Max	25.00	1.00×10^{12}	1.16×10^9	100

3.3. Robust model comparison

To visually compare the performance of the four tested supervised ML model (RFR, BR, XGBR, and DTR), we visualized the relationship between the actual and predicted algicidal ratios (Fig. 1) and plotted their residuals (Fig. 2).

The actual versus predicted algicidal ratios indicate superior performance for the RFR, BR, and XGBR models. Specifically, the RFR, BR, and XGBR models demonstrated high accuracy rates of 93 %, 91 %, and 90 %, respectively. Conversely, the DTR model exhibited a high accuracy on the training dataset (98 %) but its performance level dropped significantly (50 %) on the testing dataset, signifying potential overfitting. Previous studies have also revealed varying performance of these models (Mustafa et al., 2024; Xu et al., 2024). The high level of accuracy of the RFR model can be ascribed to its model structure, which effectually handled large, missing, and outlier data (Haggerty et al., 2023).

The enhanced performance of the BR model was typically due to its ability to reduce variance by averaging predictions from various base models (Haggerty et al., 2023). The performance of the XGBR model stems from its enhanced gradient boosting framework, which is optimized for speed, is well suited for large datasets, and employs regularization to reduce overfitting by setting an error reduction threshold.

This model has consistently outperformed other machine learning models across a variety of datasets (Ransom et al., 2022). In contrast, the overfitting observed in the DTR model is likely due to reliance on spurious correlations between variables, leading to suboptimal partitioning and reduced generalization on unseen data (Zhang et al., 2024a).

The random distribution of residuals around the zero line in Fig. 2 further validated the results, indicating that the RFR, BR, and XGBR models reliably captured underlying spatial patterns from data, thereby confirming the accuracy of their predictions (Jeong et al., 2024). The evaluation metrics for tested ML models across training, validation, and testing are presented in Table 2.

3.4. Quantitative sensitivity pattern analysis

Previous studies have investigated the effects of culture time, microorganism concentration, and algal biomass on the algicidal process against HABs primarily through experimental approaches, providing mainly qualitative insights (Lee et al., 2024). In contrast, our study distinguishes itself by adopting a data-driven approach, which offers the potential to uncover hidden patterns and complex, nonlinear relationships within multisource experimental datasets (Yan et al., 2024).

While experimental research has focused on assessing the impacts of specific conditions, our application of ICE plots for sensitivity pattern analysis enables a more nuanced understanding of how features such as culture time, concentration, and algal biomass influence the algicidal activity of microorganisms against HABs. This quantitative sensitivity analysis provides clear insights into the individual effects of each variable, a level of detail that traditional experimental methods struggle to achieve when multiple features interact concurrently (Fu et al., 2024).

The findings from sensitivity pattern analysis are shown in Fig. 3. ICE plots in Fig. 3a revealed that all four tested models exhibited sensitivity to changes in culture time during initial stages. This suggests that the impact of culture time on algicidal ratio diminishes as culture time increases. Fig. 3b illustrates the effect of concentration on the algicidal ratio across all models. Concentration has a significant impact on the algicidal ratio, particularly at lower levels. As concentration increases, the sensitivity of the models decreases, resulting in more stable predictions at higher concentrations.

During initial stages, when algal biomass is low, as shown in Fig. 3c, the algicidal ratio is initially high but decreases sharply as algal biomass increases. This underscores the critical role of the initial conditions in algicidal processes, emphasizing the need for precise management of factors like concentration and culture time to optimize microbial treatments for HAB management.

After the initial drop in algicidal ratio, models indicated stabilization or a slight recovery in algicidal effects at higher biomass levels. This

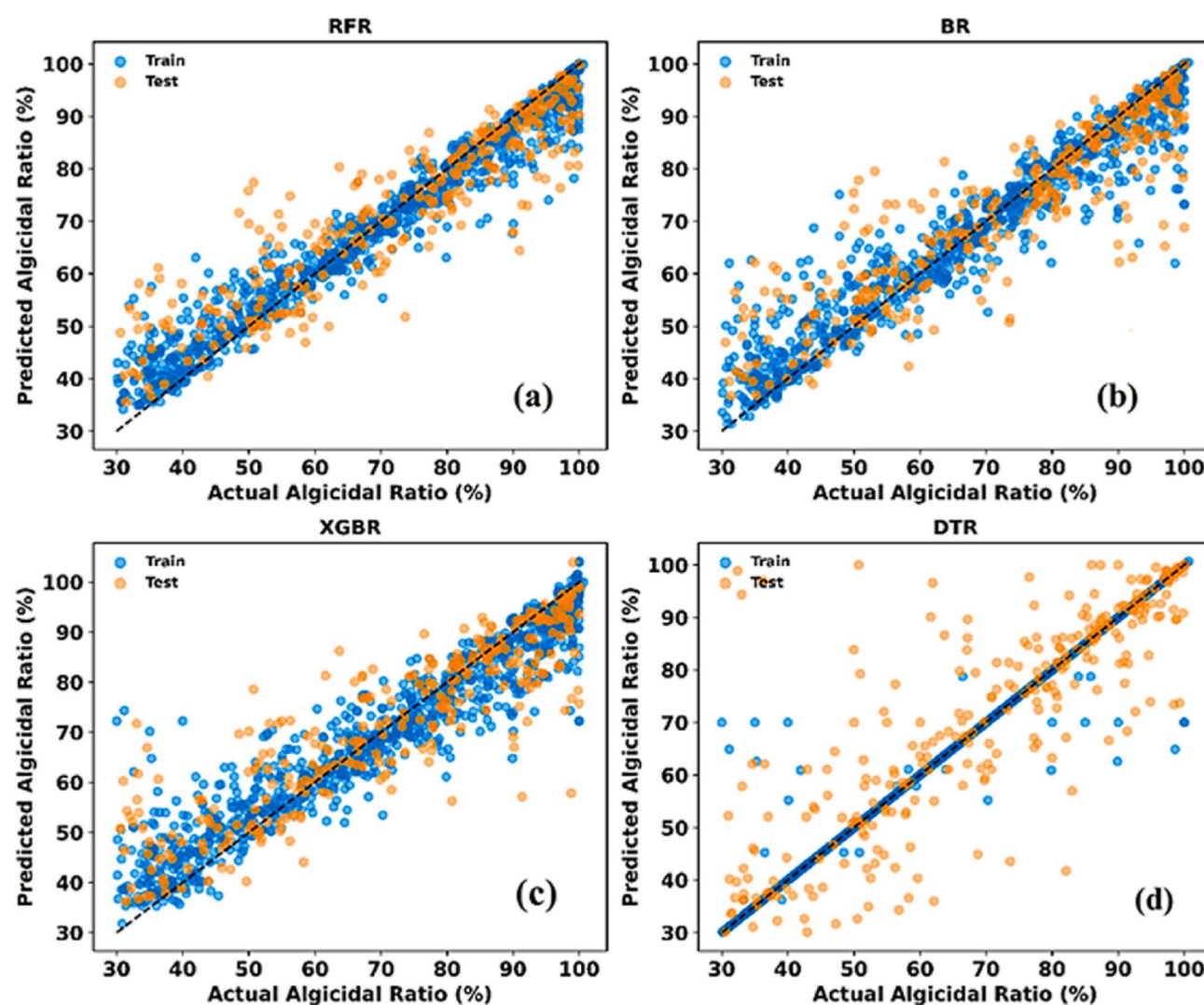


Fig. 1. Actual versus predicted algicidal ratios (%) for the (a) RFR, (b) BR, (c) XGBR, and (d) DTR models.

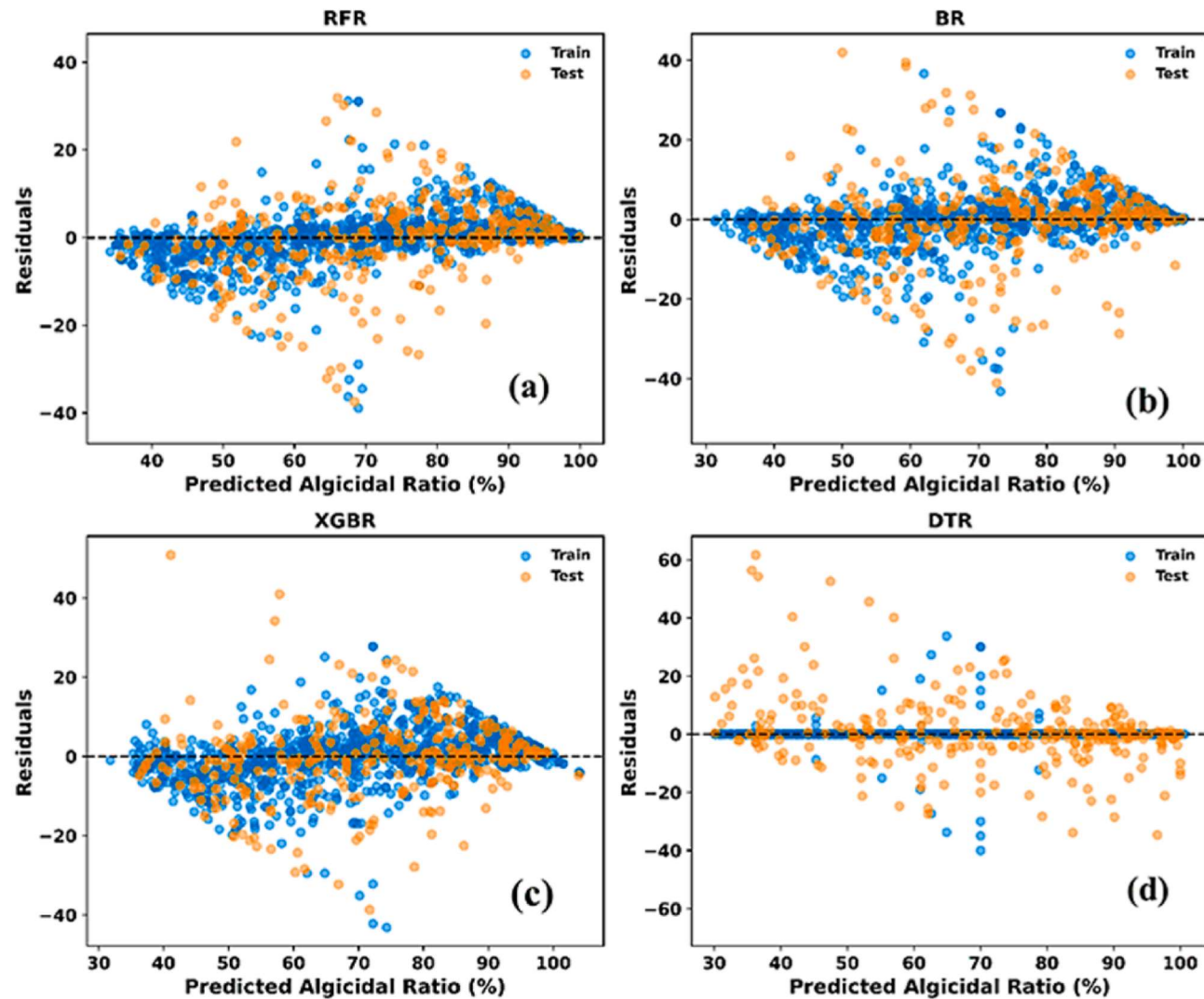


Fig. 2. Residual analysis plots of actual versus predicted algicidal ratio (%) for the (a) RFR, (b) BR, (c) XGBR, and (d) DTR models.

Table 2

Performance metrics of tested machine learning (ML) models in predicting algicidal ratios.

Metrics	RFR	BR	XGBR	DTR
Train R^2	0.93	0.91	0.90	0.97
Validation	0.88	0.87	0.82	0.62
Test	0.87	0.85	0.80	0.60
Train MAE	0.049	0.053	0.0645	0.0076
Validation MAE	0.126	0.130	0.1294	0.1498
Test MAE	0.110	0.113	0.1060	0.1309
Train RMSE	0.0811	0.091	0.0948	0.0516
Validation RMSE	0.1875	0.196	0.1867	0.2451
Test RMSE	0.1657	0.171	0.1567	0.2132

trend calls for further investigation to better interpret fundamental mechanisms that influence the efficiency of microorganisms at varying algal biomass levels.

3.5. Threshold estimation in microbial–algal interaction dynamics

Partial dependence plots (PDPs) for the tested models reveal nonlinear relationships between the response and explanatory variables. These plots display the average predicted values of the models across the range of each explanatory variables, providing insight into their respective impacts (Tong et al., 2019). The relationship between culture time and algicidal ratio, as shown in Fig. 4a showed that the threshold time for maximizing the algicidal ratio appeared to be 7–10 days. This suggests that beyond 7–10 days, further increases in time do not significantly enhance algicidal processes.

An experimental study revealed that both bacterial cultures and filtrates of strains DH74, DH77, SP31, SP44 and SP96 exhibited similar algicidal activity to *Alexandrium tamarense*. The algal cells were nearly 100 % killed by day 5 (DH74), day 2 (DH77, SP31, SP44) and day 1 (SP96) (Su et al., 2011). Additionally, strain FDHY-03 exhibited algicidal activity across a broad range of HAB-forming species, with the

largest effect observed on *Prorocentrum donghaiense*, resulting in 91.7 % mortality within 24 h (Shi et al., 2018). Our identification of a 7–10-day threshold could be due to the inclusion of broader temporal dynamics across different strains and extended time frames.

As shown in Fig. 4b, a rapid increase in partial dependence values was observed between 0.0 and 0.5×10^{10} cells/mL. Beyond this concentration level, partial dependence values across all models stabilized, indicating that relatively low initial concentrations resulted in higher algicidal activity. Increasing concentration beyond this point, up to 3.5×10^{10} cells/mL, offered diminishing returns. In the RFR and BR models, concentrations beyond 2.0×10^{10} cells/mL slightly decreased partial dependence values or the predicted algicidal ratio, suggesting that excessively high concentrations might even reduce microbial effectiveness.

The algicidal efficiency of different strains, including YZ1, CBS, GDMCC 1.867, and GDMCC 1.467, was tested against *Microcystis aeruginosa* at concentrations ranging from 1×10^6 to 1×10^8 cells/mL. The highest algicidal activity was observed at a concentration of 1×10^8 cells/mL (Liu et al., 2022a). In a separate study, a 3 % (v/v) inoculation dose of antialgal actinomycetes at a cell density of 1.2×10^8 CFU/mL was necessary for the lysis of *M. aeruginosa*, achieving an inhibition rate was 81.2 % (Kong et al., 2013).

Moreover, addition of approximately 10 % of *Enterobacter* sp. EA-1 (cell density = 2.3×10^8 CFU/mL) was sufficient for the complete degradation of algae (Lu et al., 2021). Although these microbial densities were lower, they approached the threshold concentration identified in our ML analysis (i.e., 0.5×10^{10} cells/mL) for achieving maximum algicidal ratio across diverse conditions.

As illustrated in Fig. 4c, a significant decline in partial dependence was observed between 0 and 1×10^7 cells/mL of algal biomass. After this initial decline, curves recovered and began to stabilize around 1.5×10^7 cells/mL of algal biomass. Consequently, at around 1.5×10^7 cells/mL of algal biomass, algicidal ratio showed signs of recovery and subsequently stabilized. Yu et al., 2019 focused on a limited range of algal density (1×10^5 to 1×10^7 cells/mL) and established that maximum

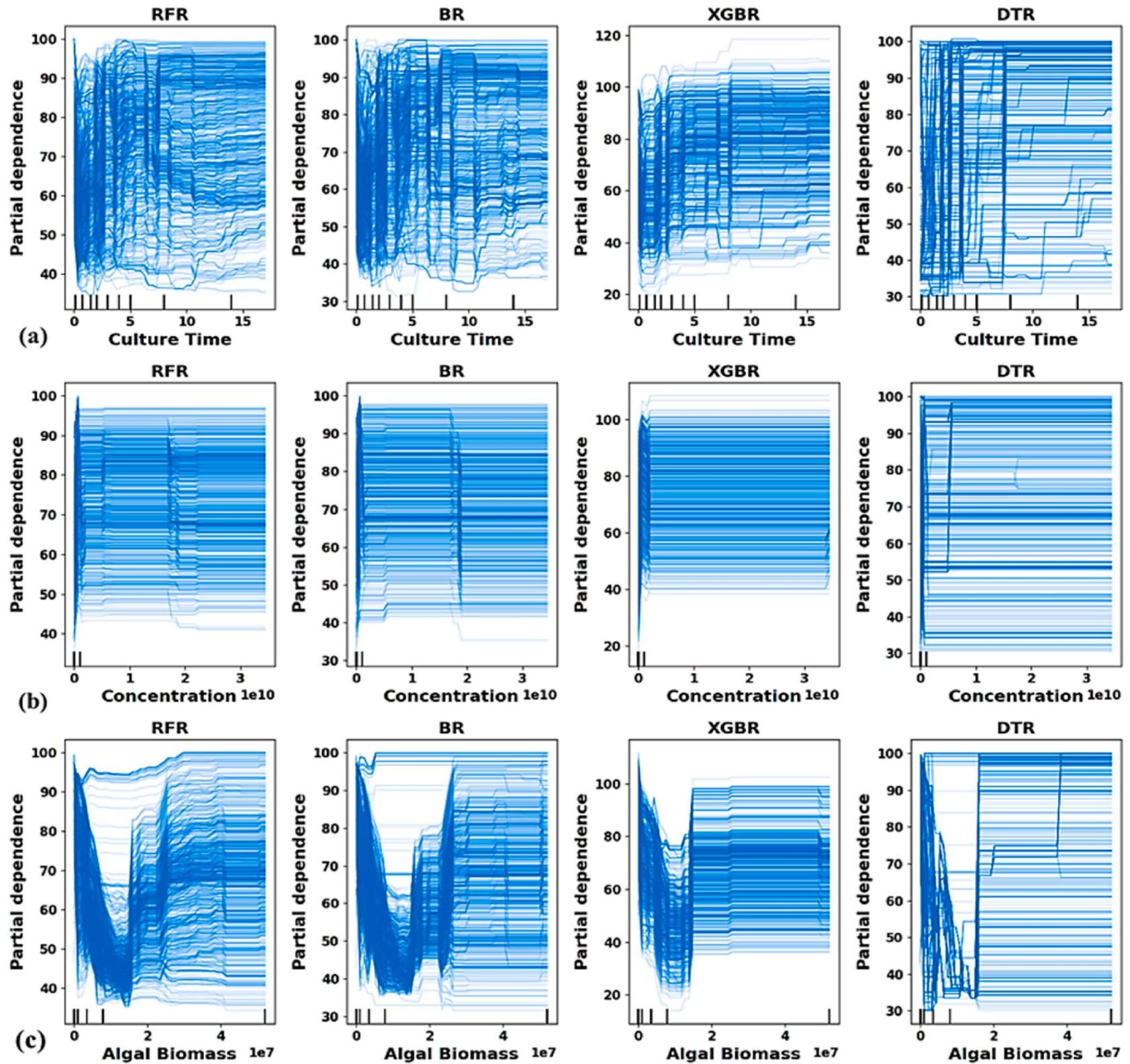


Fig. 3. Quantitative assessment of the impact of (a) culture time (days), (b) concentration (cells/mL), and (c) algal biomass (cells/mL) on the algicidal process in the presence of microorganisms against HABs using ICE plots generated by the RFR, BR, XGBR, and DTR models.

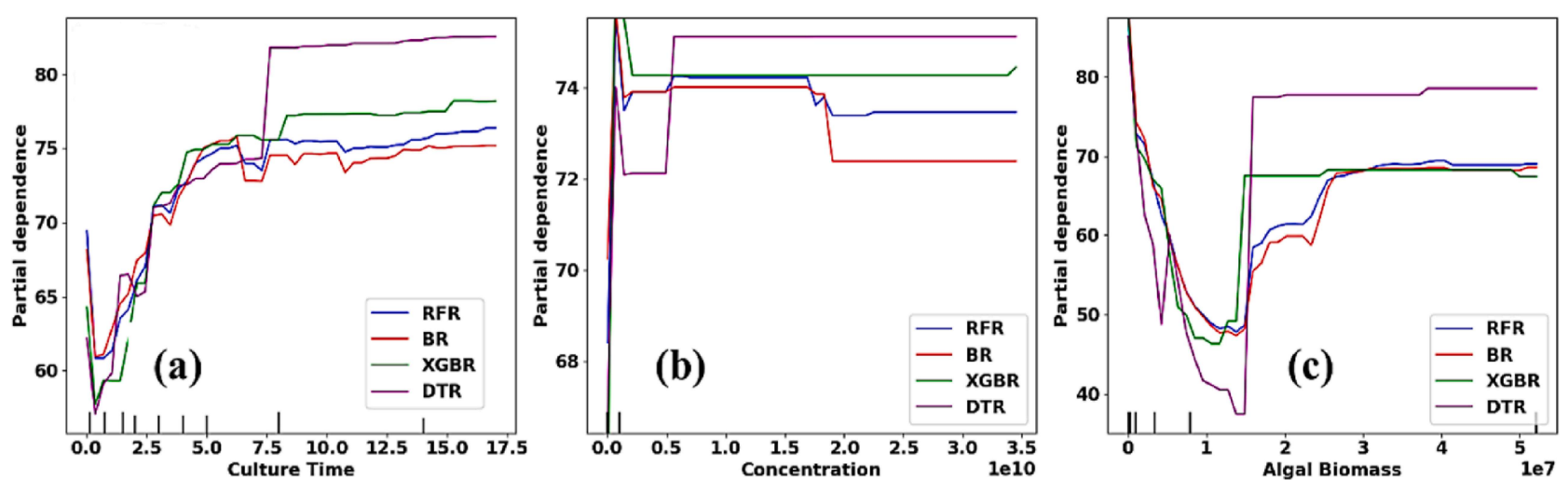


Fig. 4. Partial dependence plots of algicidal ratio (%) against explanatory variables: (a) culture time (days), (b) concentration (cells/mL), and (c) algal biomass (cells/mL) from the RFR, BR, XGBR, and DTR models.

algicidal ratios occurred at an algae density of 1×10^6 cells/mL.

However, their study did not explore effects of algae density beyond 1×10^7 cells/mL, leaving a critical gap in understanding how algicidal activity is influenced at higher algal biomass levels (Yu et al., 2019). Our

findings provide additional insights into a broader range of algal biomass densities ($0\text{--}3.5 \times 10^7$ cells/mL), suggesting that algicidal microorganisms remained effective even in environments where algal densities exceed 1×10^7 cells/mL. This extended efficiency at higher

algal biomass highlights the potential of these microorganisms as a sustainable solution for mitigating HABs at various stages of development, beyond early phases when algal biomass is low.

These identified thresholds represent the optimal ranges based on average trends observed in our collected dataset. These values are not universal and should be considered context-dependent requiring validation and adjustment for specific microbial strains, HAB species, and environmental conditions in real-world applications.

3.6. Top influential factors driving the algicidal process

Many studies have employed simple correlation analyses to assess relationships between individual environmental factors and HAB species (Xia et al., 2019). Traditional research primarily relies on mathematical equations to simulate essential physical, chemical, and biological processes (Ralston and Moore, 2020). However, due to the inherent characteristics of process-based models and the complexities of modeling nonlinear processes, traditional approaches often struggle to identify critical driving factors (Su et al., 2022).

To our knowledge, there is currently no data available that quantitatively identifies key factors influencing microbial algicidal processes against HABs using a data-driven ML-based approach. To address this gap, we analyzed average feature importance of variables across four tested models (Fig. 5). Identifying a set of significant features from a dataset is considered one of the most crucial steps in implementing data-driven techniques (Su et al., 2022). Our average feature importance analysis revealed that algal biomass and culture time consistently emerged as the two most influential factors.

Notably, these two factors have been extensively examined in traditional experiments, where they have been frequently identified as key drivers of interactions between microbes and HABs (Su et al., 2007). The strong alignment between our machine learning findings and previous experimental studies further underscores the importance of algal biomass and culture time in mitigating HABs. Most studies have utilized this analysis to validate their screening results (Zhang et al., 2024b).

Additionally, within our dataset of over 100 microorganisms, *Brevibacillus* emerged as the third most influential feature. This finding is consistent with prior research showing that *Brevibacillus* spp. can achieve a 100 % removal rate of *Microcystis aeruginosa* when the inoculation concentration reaches approximately 3 % (Liu et al., 2023). Microbial concentration was also identified as a critical factor, with *Enterobacter* species ranking as the fifth most important feature in the microbial community. This result aligns with studies demonstrating the strong and rapid algicidal efficacy of *Enterobacter*, which achieved complete cell lysis of *Oscillatoria* within 72 h under a high initial chlorophyll concentration and 10 % inoculation of EA-1 (cell density = 2.3×10^8 CFU/mL) (Lu et al., 2021). *Enterobacter hormaechei* F2 also displayed substantial algicidal activity against *M. aeruginosa* (Zhang et al., 2023).

Our dataset included a wide range of HAB species. Among all tested species, *Microcystis aeruginosa* and *Alexandrium tamarense* were consistently identified as top features by our models. These two species are well recognized as dominant contributors to HABs in experimental studies, further validating our model's potential to predict important ecological outcomes (Li et al., 2019; Mao et al., 2024; Yang et al., 2014).

Overall, these results provide a data-driven analysis of microbial–HAB interactions, delivering valuable insights into how various factors shape algicidal processes. While our analysis accounted for differences among microbial species and their interactions with experimental conditions, future work should further refine these models by exploring species-specific effects in greater depth and investigating additional interactions between microorganisms and environmental conditions. Such advancements will help develop more targeted and effective microbial strategies for HAB management.

4. Conclusion

This study used data-driven machine learning approaches to address key challenges encountered in the biological control of HABs using microorganisms. The aggregation of data for the diverse microbial species that are effective against HABs and the detailed descriptive

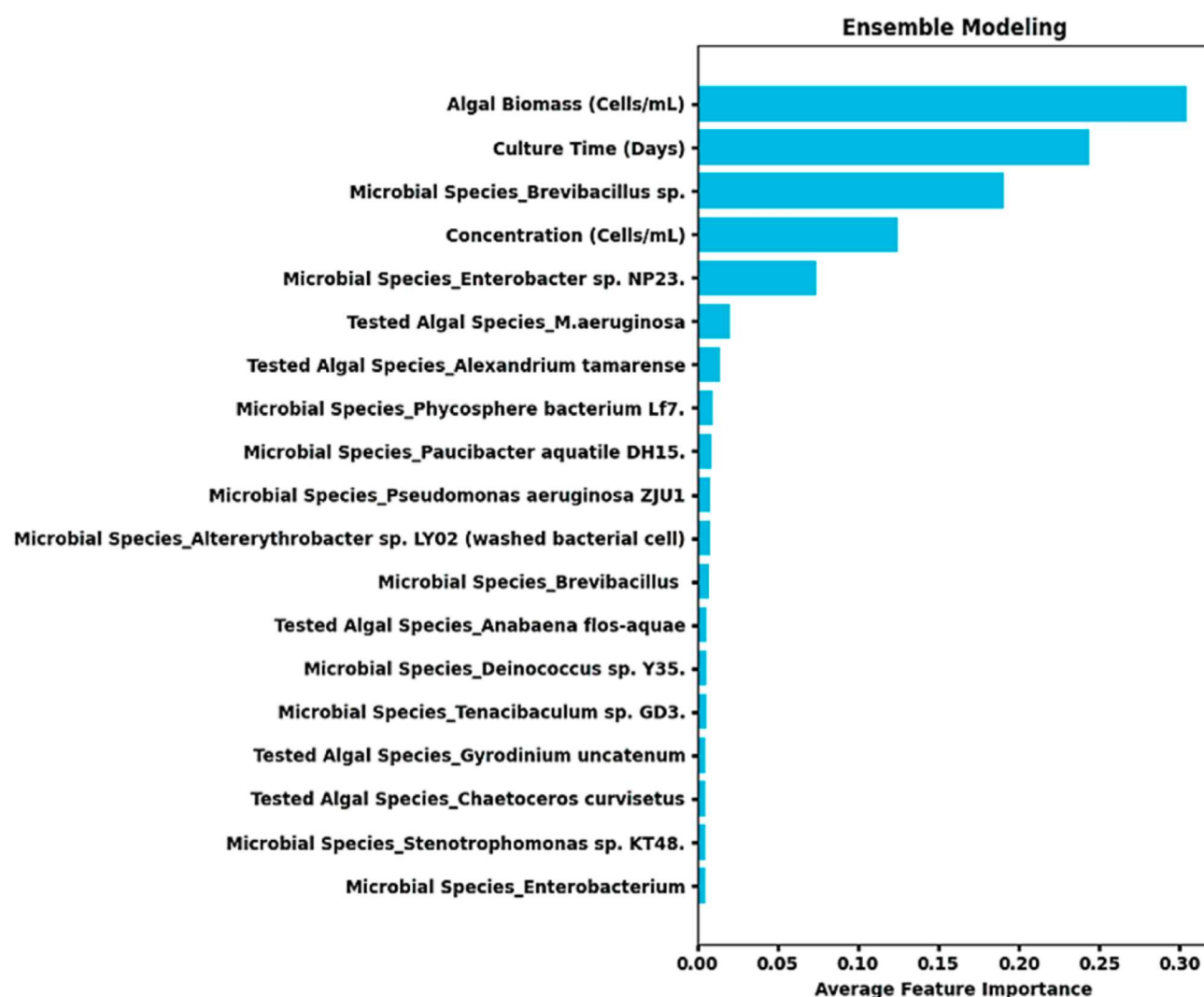


Fig. 5. The relative impact of multiple driving factors, based on the average feature importance across the four tested models, to elucidate the effectiveness of microorganisms against HABs.

statistical analysis of 2500 observations provides a comprehensive resource by offering valuable insights into well-studied and lesser-known microbial and harmful algal bloom species, as well as establishing a solid foundation for optimizing microbial-based algal treatments to achieve more consistent results.

We validated four ML models on this dataset. We found that the RFR, BR, and XGBR models demonstrated superior performance in forecasting the algicidal ratio with high accuracy and random residual distributions. However, the DTR model revealed significant overfitting, as shown by its decreased performance on the testing dataset. To further understand microbial–algal interaction dynamics, we used ICE plots to explore sensitivity pattern analysis.

Our results showed that culture time, concentration, and algal biomass significantly influenced the algicidal processes of microorganisms against HABs. Specifically, the impact of culture time diminished over its duration, concentration sensitivity decreased at higher levels, and algicidal ratio sharply declined with increasing algal biomass before reaching equilibrium. We further investigated the threshold identification to maximize algicidal ratios, highlighting the potential of microorganisms for sustained mitigation of HABs across varying culture times, concentration levels, and algal biomasses.

A feature importance analysis revealed key factors influencing algicidal processes, providing an intuitive understanding the marginal effects of each factor had on algicidal ratios. These insights offer a deeper and more systematic understanding of how microorganisms affect HABs. Furthermore, this research underscores the value of machine learning methods in predicting the effects of biological agents on HABs.

Going forward, enhancing HAB mitigation will benefit from integrating additional environmental variables (e.g., temperature, light, pH, nutrients) into machine learning models. Moreover, species-specific models and field validations are needed to ensure the effectiveness of predicted thresholds under real-world conditions. A more granular approach, including interaction terms or dedicated species -level models, presents a promising direction for future research.

CRedit authorship contribution statement

Zobia Khatoon: Writing – original draft. **Suiliang Huang:** Supervision.

Ethical statement

This study uses publicly available, peer reviewed data from published studies. No primary data involving human or animal subjects were collected. All sources have been properly cited, and the research adheres to ethical guidelines for data usage and scientific integrity.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.watbs.2025.100503>.

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